

CHANDIGARH UNIVERSITY APEX INSTITUTE OF TECHNOLOGY



EXPERIMENT 9

**Subject: Advanced Database Management
System**

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Question 1:

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Creating the procedure:

Query Query History

```
235 INSERT INTO employee_info (name, gender, salary, city)
236 VALUES
237 ('Alok', 'Male', 50000.00, 'Delhi'),
238 ('Priya', 'Male', 60000.00, 'Mumbai'),
239 ('Rajesh', 'Female', 45000.00, 'Bangalore'),
240 ('Sneha', 'Male', 55000.00, 'Chennai'),
241 ('Anil', 'Male', 52000.00, 'Hyderabad'),
242 ('Sunita', 'Female', 48000.00, 'Kolkata'),
243 ('Vijay', 'Male', 47000.00, 'Pune'),
244 ('Ritu', 'Male', 62000.00, 'Ahmedabad'),
245 ('Amit', 'Female', 51000.00, 'Jaipur');
246
247
248 CREATE OR REPLACE PROCEDURE SP_PROC(IN p_GENDER varchar, out p_tot_cnt int)
249 LANGUAGE PLPGSQL
250 as
251 $$
252 DECLARE
253 begin
254 select count(id) into p_tot_cnt from employee_info
255 where gender=p_GENDER;
256 Raise notice 'Total count of % is %', p_GENDER, p_tot_cnt;
257
258 END;
259 $$;
260
```

Data Output Messages Notifications

CREATE PROCEDURE

Query returned successfully in 287 msec.

Calling the procedure:

260
261 call SP_PROC('Male', NULL);
262
263
264
265 |

Data Output Messages Notifications



Showing rows: 1 to 1 Page No: 1 of 1

p_tot_cnt	integer
1	6

Question 2:

Problem Statement [Hard]

- An e-commerce company maintains product stock information in the products table and stores completed orders in the sales table. Create a PostgreSQL stored procedure that accepts a **product name and required quantity as input parameters** and processes the purchase only when sufficient stock is available.
- The procedure should verify the available quantity of the requested product. If sufficient stock exists, it should retrieve the product code and price, calculate the total sale amount, insert a new order into the sales table, reduce the available stock, and increase the total quantity sold. If sufficient stock is not available, the procedure should not modify any data and should display an appropriate notification message.
- The total sale amount should be calculated as:
- $\text{Sale Price} = \text{Product Price} \times \text{Quantity Ordered}$

Create the procedure:

```
Query History
301
302 CREATE OR REPLACE PROCEDURE pr_buy_products(
303     IN p_product_name VARCHAR,
304     IN p_quantity INT
305 )
306 LANGUAGE plpgsql
307 AS $$
308 DECLARE
309     v_product_code VARCHAR(20);
310     v_price FLOAT;
311     v_count INT;
312 BEGIN
313
314     SELECT COUNT(*)
315     INTO v_count
316     FROM products
317     WHERE product_name = p_product_name
318     AND quantity_remaining >= p_quantity;
319     IF v_count > 0 THEN
320         SELECT product_code, price
321         INTO v_product_code, v_price
322         FROM products
323         WHERE product_name = p_product_name;
324         INSERT INTO sales (order_date, product_code, quantity_ordered, sale_price)
325         VALUES (CURRENT_DATE, v_product_code, p_quantity, (v_price * p_quantity));
326         UPDATE products
327         SET quantity_remaining = quantity_remaining - p_quantity,
328             quantity_sold = quantity_sold + p_quantity
329         WHERE product_code = v_product_code;
330
331         RAISE NOTICE 'PRODUCT SOLD..! Order placed successfully for % unit(s) of %.', p_quantity, p_product_name;
332
333     ELSE
334         RAISE NOTICE 'INSUFFICIENT QUANTITY..! Order cannot be processed for % unit(s) of %.', p_quantity, p_product_name;
335     END IF;
336 END;
337 $$;
```

Calling the procedure:

```
337
338
339 call pr_buy_products('Samsung Galaxy S23 Ultra', 3);
340
341
```

Data Output Messages Notifications

NOTICE: PRODUCT SOLD..! Order placed successfully for 3 unit(s) of Samsung Galaxy S23 Ultra.
CALL

Query returned successfully in 49 msec.